1. Nuclear fusion reaction is also called thermonuclear reaction. Why?

Nuclear fusion reaction is called thermonuclear reaction because it occurs only at **extremely high temperature (millions of Kelvin)**. The high temperature provides sufficient kinetic energy to overcome the repulsive force between nuclei, allowing them to fuse together.

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2. Explain the harms caused by nuclear radiations or radioactivity on human health.

Nuclear radiations are highly harmful to human health. They can ionize body cells and damage biological tissues. Their major effects include mutation in DNA, cancer, skin burns, destruction of bone marrow, and damage to internal organs. High exposure may also lead to sterility or even death.

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3. Distinguish between natural radioactivity and artificial radioactivity.

Natural Radioactivity	Artificial Radioactivity
Occurs naturally in nature.	Produced by artificial nuclear reactions.
Unstable nuclei exist in nature.	Stable nuclei are made radioactive by bombardment.
Example: Uranium, Radon.	Example: Phosphorus-30, Cobalt-60.

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4. Write down any two uses of radioisotopes in medical field.

1. Radioisotopes are used in **diagnosis of diseases**, such as detecting tumors and organ functioning.
2. They are used in **radiotherapy** to treat cancer by destroying cancerous cells.

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5. Differentiate between Nuclear reaction and Chemical reaction.

Nuclear Reaction	Chemical Reaction
Occurs in nucleus of atom.	Occurs in outer electrons.
Involves change in nucleus.	No change in nucleus.
Produces large energy.	Produces small energy.
Example: Fission, fusion.	Example: Burning, rusting.

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6. What is radioactivity? Give its units.

Radioactivity is the phenomenon in which unstable atomic nuclei **spontaneously disintegrate** and emit radiation such as alpha, beta, and gamma rays.

👉 **Units of radioactivity:**

- Curie (Ci) = 3.7×10^{10} disintegrations per second
- Becquerel (Bq) = 1 disintegration per second
- Rutherford (Rd) = 10^{06} disintegrations per second

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7. What is meant by artificial radioactivity? Give an example of it.

Artificial radioactivity is the phenomenon in which **stable nuclei are made radioactive by artificial nuclear bombardment**.

👉 **Example:**
Aluminium-27 when bombarded with alpha particles produces radioactive phosphorus-30:

27/13 Al + 4/2He → 30/15 P + 1/0n